Problem 10.32

Consider the following problem: Invest 1 at a compound interest rate i for integral periods and at a simple interest rate i for fractional periods.

(a) Show that $i_k = i$ for $k = 1, 2, \ldots, n$.

(b) Show that $\delta_{t_1} \neq \delta_{t_2}$ for $k < t_1 < t_2 < k+1$, where $k = 1, 2, \ldots, n-1$.

Thus, a constant effective rate of interest i does not necessarily imply a constant force of interest.

Solution

For $k \leq t < k+1$, where k is a nonnegative integer, the investment has accumulated at compound interest for k complete periods and at simple interest during the fractional period $t-k$. Therefore, its accumulation function is

$$a(t) = (1+i)^k \big[1 + i(t-k)\big].$$

At an integer time k this gives

$$a(k) = (1+i)^k.$$

Part (a).

The effective rate of interest during the kth period is

$$\begin{aligned} i_k &= \frac{a(k)}{a(k-1)} - 1 \\ &= \frac{(1+i)^k}{(1+i)^{k-1}} - 1 \\ &= (1+i) - 1 = i. \end{aligned}$$

Consequently,

$$\boxed{i_k = i, \qquad k = 1, 2, \ldots, n.}$$

Part (b).

Within the interval $k < t < k+1$, differentiate the accumulation function:

$$a'(t) = i(1+i)^k.$$

The force of interest at time t is therefore

$$\begin{aligned} \delta_t &= \frac{a'(t)}{a(t)} \\ &= \frac{i(1+i)^k}{(1+i)^k[1+i(t-k)]} \\ &= \frac{i}{1+i(t-k)}. \end{aligned}$$

If $k < t_1 < t_2 < k+1$ and $i > 0$, then

$$1 + i(t_1 - k) < 1 + i(t_2 - k).$$

It follows that

$$\delta_{t_1} = \frac{i}{1 + i(t_1 - k)} > \frac{i}{1 + i(t_2 - k)} = \delta_{t_2}.$$

Hence,

$$\boxed{\delta_{t_1} \neq \delta_{t_2}.}$$

Although every complete period has the same effective interest rate i, the force of interest decreases throughout each fractional period and jumps back to i at the beginning of the next period. Therefore, a constant effective rate does not necessarily produce a constant force of interest.